



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



УНИВЕРЗИТЕТ У БЕОГРАДУ
UNIVERSITY OF BELGRADE

Quasi Periodic Eruptions

ACTIVE GALACTIC NUCLEI

Fritz Ali Agildere

April 8, 2026



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- $10^{5.5-6.5} M_{\odot}$ Mass Systems



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- Missing Optical Broad Lines (No BLR)
- Narrow Lines Present (Past AGN Like Activity)
- Extended Emission Line Regions (Short Lived SMBH Flare)

Models

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 - Repeating Partial Tidal Disruptions (pTDEs)

Models

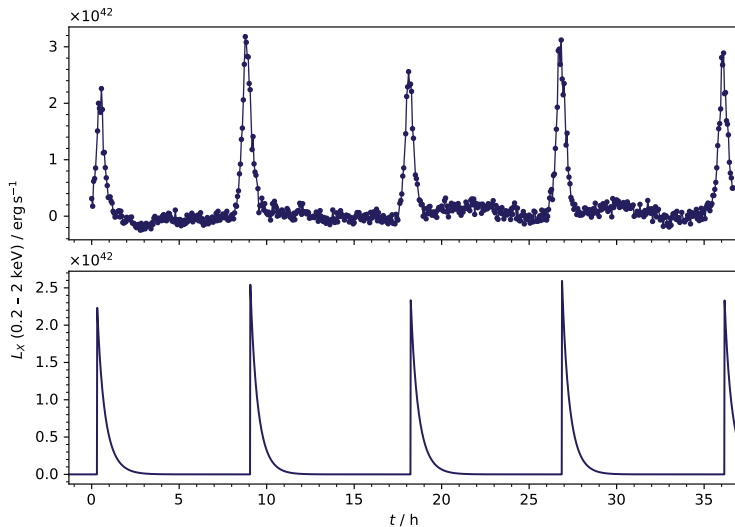
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 - Extreme Mass Ratio Inspiral (SMO/BH EMRI)

Source: GSN-069

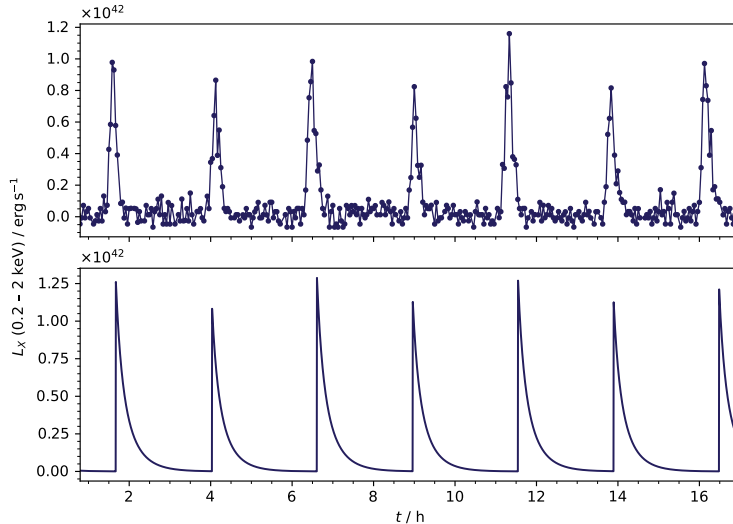
Source: GSN-069



Examples

Source: eRO-QPE2

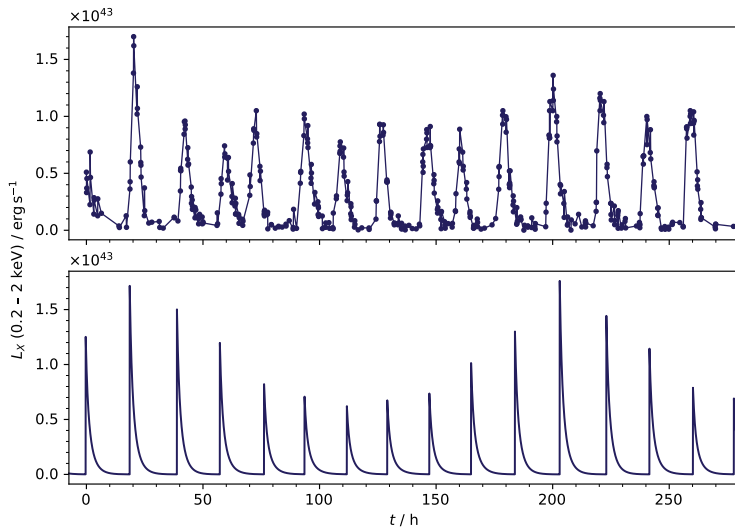
Source: eRO-QPE2



Examples

Source: eRO-QPE1

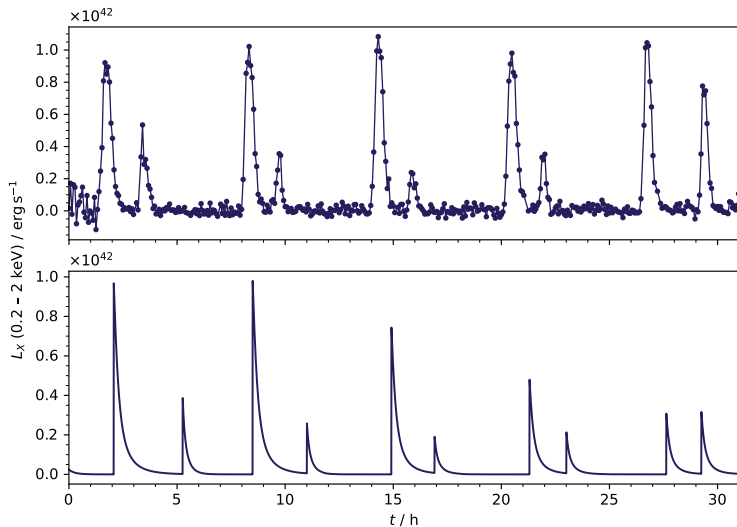
Source: eRO-QPE1



Examples

Source: RX-J1302

Source: RX-J1302



Examples



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Outlook

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- Longer Term XMM-Newton / NICER Observations (Exclude Models)

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- Precision Cosmology Measurements

Thank You!

- E. Kara & J. García, *Supermassive Black Holes in X-Rays: Standard Accretion to Extreme Transients* (2025) arXiv:2503.22791 [astro-ph.HE]
- A. Franchini et. al., *Quasi Periodic Eruptions from Impacts with Rigidly Precessing Accretion Disc in Extreme Mass Ratio Inspiral Systems* (2023) A&A Volume 675 Article 100

Resources on GitHub: [here](#)